Complete Generative AI Series with Python & LangChain - YouTube

By TechSimPlus Learnings | Prateek Mishra

Total Videos: 45 to 50 **Duration:** 40+ Hours **Projects:** 10+ Amazing Projects **Support:** 1-to-1 Doubt Solving

What You'll Learn

- ☐ Large Language Models (LLMs) fundamentals
- ☐ LangChain from basics to advanced
- ☐ Build AI Agents and Autonomous Systems
- ☐ RAG (Retrieval Augmented Generation) Systems
- ☐ AWS Bedrock for Enterprise AI
- ☐ LangGraph for Complex Workflows
- ☐ MCP Servers & Production Deployment

Prerequisites

- ☐ Basic Python Programming Knowledge
- ☐ Laptop with 8GB RAM
- ☐ Enthusiasm to Learn! 🔥

🎯 MODULE 1: FUNDAMENTALS

1. Series Introduction

What is GenAI? | Complete Roadmap | What We'll Build

- ☑ ~~Introduction to the series~~
- ☑ ~~Complete content roadmap~~
- ☑ ~~What makes this different~~
- ☑ ~~Projects preview~~
- ☑ ~~1-to-1 Support details~~

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2. What is LLM and How It Works?

Large Language Models Explained | GenAI Fundamentals

- ☑ ~~What are Large Language Models?~~
- ☑ ~~How LLMs are trained~~
- ☑ ~~Understanding transformers~~
- ☑ ~~Popular LLMs: GPT, Claude, Gemini, Llama~~
- ☑ ~~LLM capabilities and limitations~~

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3. Tokenization & Building Custom Tokenizer

How LLMs Understand Text | Tokenization Deep Dive

- ☑ ~~What is tokenization?~~
- ☑ ~~Importance of tokenization ?~~
- ☑ ~~Build your own tokenizer.~~
- ☑ ~~Token limits and costs~~

🔧 MODULE 2: SETUP & FIRST STEPS

4. Complete API Setup Guide

OpenAI, Google Gemini, Groq, Anthropic Claude

- ☑ ~~Create OpenAI account & get API key~~
- ☑ ~~Setup Google Gemini API~~
- ☑ ~~Get Groq API access~~
- ☑ ~~Anthropic Claude setup~~
- ☑ ~~HuggingFace account setup~~
- ☑ ~~Environment variables & security~~
- ☑ ~~API cost management~~

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6. Getting Started with LangChain & OpenAI

Your First LangChain Application

- ☑ ~~What is LangChain?~~
- ☑ ~~LangChain installation~~
- ☑ ~~Basic LangChain structure~~
- ☑ ~~Connect OpenAI with LangChain~~
- ☑ ~~Your first LLM call~~

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7. Prompt Templates & Chains

LangChain Expression Language (LCEL)

- ☑ ~~What are Prompt Templates?~~
- ☑ ~~Dynamic prompts with variables~~
- ☑ ~~LangChain chains~~
- ☑ ~~LCEL (LangChain Expression Language)~~
- ☑ ~~Chaining multiple operations~~
- ☑ ~~String output parsers~~

💬 MODULE 3: BUILDING CHATBOTS

7. Build AI Chatbot with LangChain & OpenAI

End-to-End QnA ChatBot with Streamlit UI

- ☑ ~~Design chatbot architecture~~
- ☑ ~~LangChain + OpenAI integration~~
- ☑ ~~Build Streamlit interface~~
- ☑ ~~Handle user input~~
- ☑ ~~Display responses~~

🎯 **Project:** QnA Chatbot with OpenAI

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8. Add Memory to AI Chatbot

LangChain Message History | Conversation Memory

- ☑ ~~Why memory matters~~
- ☑ ~~ConversationBufferMemory~~
- ☑ ~~Message history~~
- ☑ ~~Session management~~
- ☑ ~~Context retention~~
- ☑ ~~Memory implementation~~

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9. Run LLMs Locally FREE with Ollama

Ollama + LangChain | Local AI Models

- ☑ ~~What is Ollama?~~
- ☑ ~~Install Ollama~~
- ☑ ~~Download local models (Llama, Mistral, Phi)~~
- ☑ ~~Connect Ollama with LangChain~~
- ☑ ~~Build offline chatbot~~
- ☑ ~~Pros and cons of local LLMs~~

🎯 **Project:** Offline Chatbot with Ollama

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10. Groq AI - Fastest LLM Inference

Groq + LangChain Integration

- ☑ ~~What is Groq?~~
- ☑ ~~Why Groq is fastest~~
- ☑ ~~Setup Groq API~~
- ☑ ~~LangChain + Groq integration~~
- ☑ ~~Speed benchmarks~~

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11. Build Chatbot with Groq & Streamlit

Ultra-Fast AI Chatbot with Message History

- ☑ ~~Complete chatbot with Groq~~
- ☑ ~~Add message history~~

- ☑ ~~Streamlit UI design~~
- ☑ ~~Session state management~~
- ☑ ~~Deploy chatbot~~

🎯 **Project:** Groq Chatbot

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12. Build Chatbot with Google Gemini

Google Gemini + LangChain + Streamlit

- ☑ ~~Setup Google Gemini~~
- ☑ ~~LangChain integration~~
- ☑ ~~Build chatbot interface~~
- ☑ ~~Add memory~~
- ☑ ~~Handle responses~~

🎯 **Project:** Gemini Chatbot

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13. Build Chatbot with HuggingFace Models

Free Open-Source LLMs | HuggingFace + LangChain

- ☑ ~~HuggingFace Hub introduction~~
- ☑ ~~Access free models~~
- ☑ ~~Setup HuggingFace API~~
- ☑ ~~Integrate with LangChain~~
- ☑ ~~Build QnA chatbot~~
- ☑ ~~Model selection tips~~

🎯 **Project:** HuggingFace Chatbot

🤖 MODULE 4: AI AGENTS

14. What are AI Agents?

Autonomous AI | ReAct Framework Explained

- ☑ ~~What makes an agent autonomous?~~
- ☑ ~~Agent vs Chatbot~~
- ☑ ~~ReAct (Reasoning + Acting) pattern~~
- ☑ ~~Agent architecture~~
- ☑ ~~When to use agents~~
- ☑ ~~Real-world examples~~

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16. Build Your First AI Agent

LangChain + OpenAI Function Calling

- ☑ ~~Agent fundamentals~~
- ☑ ~~Function calling explained~~
- ☑ ~~Create simple tools~~
- ☑ ~~Agent executor~~
- ☑ ~~Tool selection logic~~
- ☑ ~~Error handling~~

🎯 **Project:** Basic Calculator Agent

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17. AI Agent Tools & Function Calling

Create Custom Tools for Your Agents

- ☑ ~~What are agent tools?~~
- ☑ ~~Built-in LangChain tools~~
- ☑ ~~Create custom tools~~
- ☑ ~~Tool descriptions~~
- ☑ ~~Multiple tool handling~~
- ☑ ~~Best practices~~

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18. Build Google Search Agent

AI Agent with Web Search | LangChain + Gemini

- ☑ ~~Setup search API~~
- ☑ ~~Create search tool~~
- ☑ ~~Build search agent~~
- ☑ ~~Parse search results~~
- ☑ ~~Combine multiple sources~~
- ☑ ~~Streamlit interface~~

🎯 **Project:** Google Search Agent

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19. Build SQL Agent

Chat with PostgreSQL Database

- ☑ ~~Setup PostgreSQL~~
- ☑ ~~SQL agent fundamentals~~
- ☑ ~~Database connection~~
- ☑ ~~Natural language to SQL~~
- ☑ ~~Query execution~~
- ☑ ~~Security considerations~~

🎯 **Project:** SQL Agent

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20. Build GitHub AI Agent - Self

Automate Repository Management

- ☐ GitHub API setup
- ☐ Create GitHub tools
- ☐ Repository operations
- ☐ Issue management
- ☐ PR automation
- ☐ Code search

🎯 **Project:** GitHub Automation Agent

21. Build Email AI Agent - LangGraph One Shot

Auto-Read & Send Emails

- ☐ Gmail API setup
- ☐ Email reading agent
- ☐ Email sending automation
- ☐ Smart filtering
- ☐ Response generation
- ☐ Security & permissions

🎯 **Project:** Email Assistant Agent

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22. Build Jira AI Agent - Not Needed

Automate Project Management

- ☐ Jira API integration
- ☐ Create/update tickets
- ☐ Project tracking
- ☐ Sprint management
- ☐ Automated workflows
- ☐ Team productivity

🎯 **Project:** Jira Automation Agent

📚 MODULE 5: RAG SYSTEMS

22. What is RAG?

Retrieval Augmented Generation Explained

- ☑ ~~What is RAG?~~
- ☑ ~~Why RAG is important~~
- ☑ ~~RAG architecture~~
- ☑ ~~RAG vs other approaches~~
- ☑ ~~Use cases~~
- ☑ ~~Limitations~~

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23. Vector Embeddings & Vector Databases

Understanding Embeddings | Pinecone, ChromaDB, FAISS

- [ ] What are embeddings?
- [ ] How embeddings capture meaning
- [ ] Vector databases explained
- [ ] Similarity search
- [ ] Popular vector DBs comparison

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24. Document Loaders

Load Data from Multiple Sources

- [x] ~~PDF loader~~
- [x] ~~DOCX loader~~
- [x] ~~CSV loader~~
- [x] ~~Web scraping~~
- [x] ~~API data loading~~
- [x] ~~Custom loaders~~

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25. Text Chunking Strategies

RecursiveCharacterTextSplitter Tutorial

- [ ] Why chunking matters
- [ ] Chunk size optimization
- [ ] RecursiveCharacterTextSplitter
- [ ] Overlap strategies
- [ ] Best practices

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26. Vector Embeddings & Storage

Pinecone, ChromaDB, FAISS Tutorial

- [ ] Generate embeddings

- [ ] Choose embedding model
- [ ] Vector database setup
- [ ] Store embeddings
- [ ] Index management

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27. Build RAG Chatbot with PDFs

Chat with PDF Documents + Memory

- [ ] Complete RAG pipeline
- [ ] PDF processing
- [ ] Vector storage
- [ ] Retrieval logic
- [ ] Conversation memory
- [ ] Streamlit UI

🎯 **Project:** PDF Chat Application

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28. Build Website Chatbot

RAG with Web Scraping

- [ ] Website scraping
- [ ] Content extraction
- [ ] Build RAG system
- [ ] Chat with website content
- [ ] Handle dynamic content
- [ ] Deploy chatbot

🎯 **Project:** Website QnA Bot

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29. Multi-Document RAG System

Chat with Multiple PDFs

- [ ] Handle multiple documents
- [ ] Metadata filtering
- [ ] Source attribution

- [ ] Document routing
- [ ] Comparison queries
- [ ] Advanced retrieval

🎯 **Project:** Multi-Doc RAG System

🔷 MODULE 6: LANGGRAPH & ADVANCED

30. Pydantic for Data Validation

Pydantic + LangChain & LangGraph

- [ ] Pydantic basics
- [ ] Data validation
- [ ] Type safety
- [ ] Models for LangChain
- [ ] Complex schemas
- [ ] Best practices

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31. Introduction to LangGraph

Build Agentic Workflows

- [ ] What is LangGraph?
- [ ] Graphs vs Chains
- [ ] LangGraph components
- [ ] State management
- [ ] Use cases
- [ ] When to use LangGraph

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32. Build Your First LangGraph App

Simple Chatbot with LangGraph

- [ ] Create graph structure
- [ ] Define nodes
- [ ] Add edges

- [ ] State management
- [ ] Compile graph
- [ ] Run workflow

🎯 **Project:** LangGraph Chatbot

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33. Routing & Conditional Logic

LangGraph Conditional Edges

- [ ] Conditional routing
- [ ] Dynamic workflows
- [ ] State-based routing
- [ ] Multiple paths
- [ ] Error handling
- [ ] Complex logic

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34. Multi-Tool Agents in LangGraph

LangGraph ToolNode Tutorial

- [ ] ToolNode explained
- [ ] Multiple tools
- [ ] Tool selection
- [ ] Parallel execution
- [ ] Error handling
- [ ] Best practices

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35. Memory in LangGraph

Persistent State Management

- [ ] LangGraph memory
- [ ] State persistence
- [ ] Checkpoint management
- [ ] Resume workflows
- [ ] Memory implementation

- [ ] Production patterns

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36. Human-in-the-Loop Workflows

LangGraph Approval System

- [ ] Human approval nodes
- [ ] Interrupt patterns
- [ ] Resume execution
- [ ] Feedback loops
- [ ] UI integration
- [ ] Production use cases

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37. RAG with LangGraph

Advanced RAG Workflow

- [ ] RAG in LangGraph
- [ ] Multi-step retrieval
- [ ] Query refinement
- [ ] Result validation
- [ ] Complex workflows
- [ ] Performance optimization

🎯 **Project:** LangGraph RAG System

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38. AI Blog Generator

LangGraph Multi-Agent System

- [ ] Multi-agent collaboration
- [ ] Content generation workflow
- [ ] Research → Write → Edit
- [ ] Agent coordination
- [ ] Quality control
- [ ] Full automation

🎯 **Project:** AI Blog Generator

🌐 MODULE 7: APIs & DEPLOYMENT

39. FastAPI Crash Course

Build REST APIs for AI Applications

- ☐ FastAPI basics
- ☐ Routing
- ☐ Pydantic models
- ☐ Request/Response handling
- ☐ Error handling
- ☐ API documentation

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40. Build AI Chatbot REST API

FastAPI + LangChain Integration

- ☐ Design API structure
- ☐ Integrate LangChain
- ☐ Handle async operations
- ☐ Session management
- ☐ Error handling
- ☐ Test API

🎯 **Project:** Chatbot REST API

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41. Deploy Agents using LangServe & LangSmith

Production Agent Deployment | Monitoring & Debugging

- ☐ What is LangServe?
- ☐ Deploy agents with LangServe
- ☐ Introduction to LangSmith
- ☐ Tracing and debugging
- ☐ Performance monitoring
- ☐ Production best practices

🔌 MODULE 8: MCP & AWS BEDROCK

Must Study - Safety and Guardrails

42. What is MCP?

Model Context Protocol Explained | Claude MCP

- ☐ What is MCP?
- ☐ Why MCP matters
- ☐ MCP architecture
- ☐ Claude integration
- ☐ Use cases
- ☐ Future of MCP

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43. Build Custom MCP Server

Create Your Own MCP Server

- ☐ MCP server basics
- ☐ Define resources
- ☐ Create tools
- ☐ Connect to Claude
- ☐ Test MCP server
- ☐ Deploy server

🎯 **Project:** Custom MCP Server

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44. LangChain + MCP Integration

Use MCP Tools in AI Agents

- ☐ Connect MCP with LangChain
- ☐ Use MCP tools
- ☐ Agent integration
- ☐ Complex workflows
- ☐ Best practices

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45. AWS Bedrock Introduction

Setup AWS Account & Bedrock

- [ ] What is AWS Bedrock?
- [ ] Create AWS account
- [ ] Enable Bedrock
- [ ] Available models
- [ ] Pricing overview
- [ ] Security setup

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46. Build Chatbot with AWS Bedrock

LangChain + Claude 3 on Bedrock

- [ ] Connect Bedrock
- [ ] Use Claude 3
- [ ] Build chatbot
- [ ] Handle responses
- [ ] Cost optimization
- [ ] Production setup

🎯 **Project:** AWS Bedrock Chatbot

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47. Production RAG with AWS Bedrock

Enterprise RAG System

- [ ] Complete RAG pipeline
- [ ] AWS Bedrock models
- [ ] Vector storage (AWS)
- [ ] Production architecture
- [ ] Monitoring
- [ ] Scalability

🎯 **Project:** Production RAG on AWS

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48. Deploy AI with AWS Lambda

Serverless LangChain

- [ ] What is AWS Lambda?
- [ ] Serverless architecture
- [ ] Deploy LangChain
- [ ] Handle requests
- [ ] Auto-scaling
- [ ] Cost benefits

🎯 **Project:** Serverless AI API

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49. Deploy with FastAPI on AWS

Production Deployment Guide

- [ ] EC2 setup
- [ ] Deploy FastAPI
- [ ] Load balancing
- [ ] SSL/HTTPS
- [ ] Monitoring
- [ ] CI/CD pipeline

🎯 LIST OF PROJECTS

Throughout this series, you'll build these 10+ real-world projects:

1. QnA Chatbot with OpenAI

- [ ] Technologies: LangChain, OpenAI, Streamlit
- [ ] Features: Conversation, memory, user interface

2. Google Search Agent

- [ ] Technologies: LangChain, Gemini, Google Search API
- [ ] Features: Web search, information retrieval, answer generation

3. PDF Chat Application

- [ ] Technologies: LangChain, RAG, Vector DB, Streamlit
- [ ] Features: Upload PDFs, ask questions, get answers with sources

4. SQL Agent

- [ ] Technologies: LangChain, PostgreSQL, SQL
- [ ] Features: Natural language to SQL, database querying

5. GitHub Automation Agent

- [ ] Technologies: LangChain, GitHub API
- [ ] Features: Repo management, issue handling, automation

6. Email Assistant Agent

- [ ] Technologies: LangChain, Gmail API
- [ ] Features: Read emails, auto-respond, smart filtering

7. Multi-Document RAG System

- [ ] Technologies: LangChain, RAG, Vector DB
- [ ] Features: Multiple PDF handling, source attribution, comparison

8. AI Blog Generator

- [ ] Technologies: LangGraph, Multi-Agent
- [ ] Features: Research, write, edit workflow automation

9. Custom MCP Server

- [ ] Technologies: MCP Protocol, Claude
- [ ] Features: Custom tools, resource management

10. Production RAG on AWS

- [ ] Technologies: AWS Bedrock, Claude 3, LangChain
- [ ] Features: Enterprise-grade RAG, scalable architecture

Bonus Projects:

- [ ] Offline Chatbot with Ollama
- [ ] Groq Chatbot (Ultra-fast)
- [ ] Gemini Chatbot
- [ ] HuggingFace Chatbot
- [ ] Website QnA Bot

- [ ] Chatbot REST API
- [ ] LangGraph RAG System
- [ ] Serverless AI API

Ready to start your GenAI journey? Let's build something amazing together! 🚀

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Klipo Links